화학특강 (2010년 2학기)

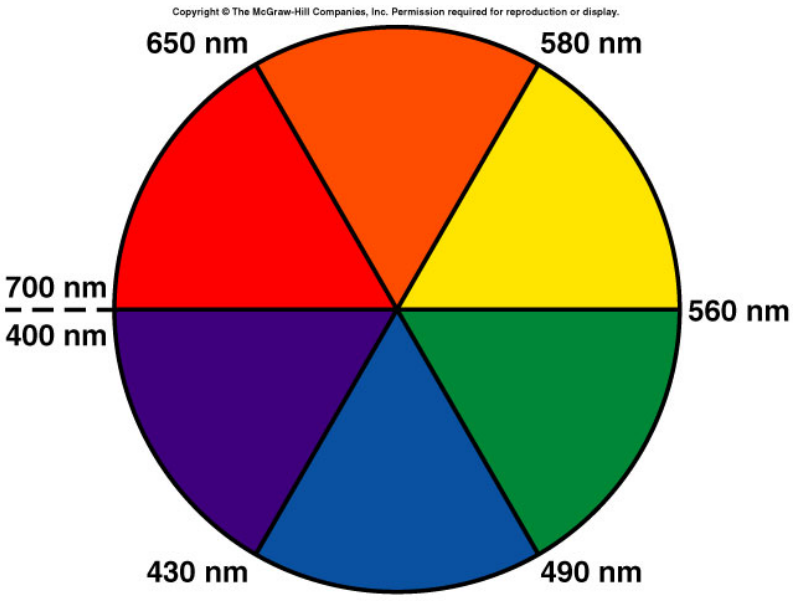
Nanotechnology와 Quantum Dots

황현석

강원대학교 화학과

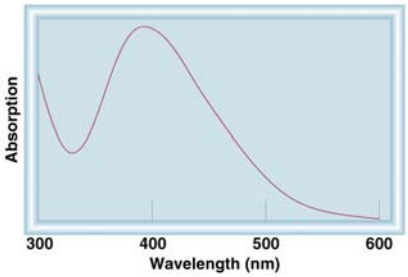
2010년 10월 25일 목요일

Complementary Colors



Absorption and Fluorescence

- Absorption spectrum of bromine



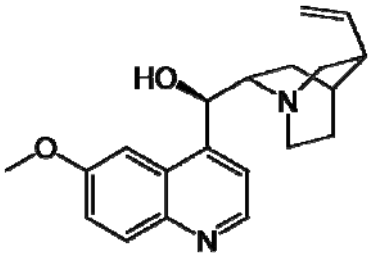


$Br_2(aq) + HCOOH(aq) \rightarrow 2Br^- + 2H^+(aq) + CO_2(g)$

Chang, 13.2

- Fluorescence of quinine



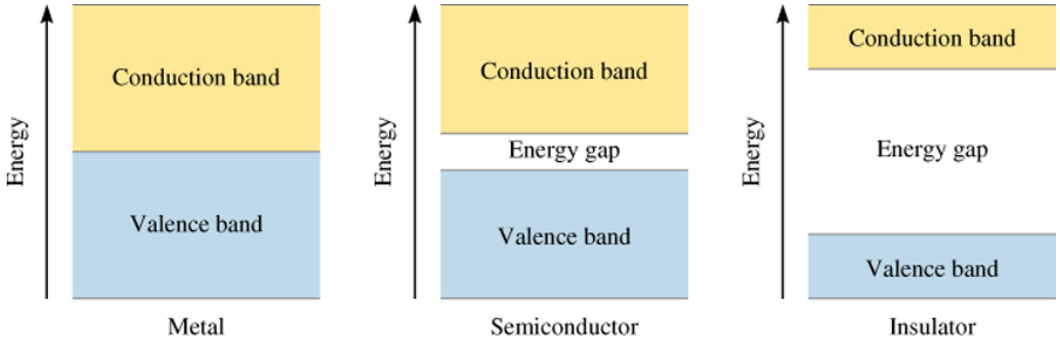


quinine

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Semiconductor

- 반도체 (semiconductor): 전도체 (conductor)와 부도체 (insulator) 사이에 존재하는 물질로서, 일정한 조건에서 도체가 되는 물질.



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Quantum Dots (QD) – an Artifical Atom

- Energy of a particle in a box $$E_n = \frac{n^2 h^2}{8 m_e L^2}$$

- Fluorescence of quantum dots

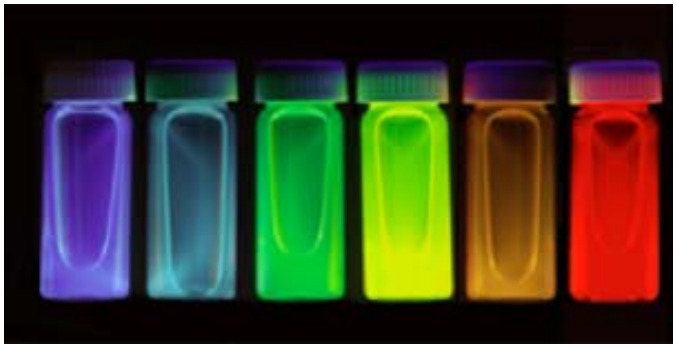
$$E_{gap}^{QD} = E_{gap}^{Bulk} + \frac{\pi^2 h^2}{2R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.8e^2}{\varepsilon R}$$

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Quantum Dots (QD) – an Artifical Atom

- Energy of a particle in a box $$E_n = \frac{n^2 h^2}{8 m_e L^2}$$

- Fluorescence of quantum dots



Colloidal CdSe quantum dots dispersed in hexane. Quantum confinement effects allow quantum-dot color to be tuned with particle size.

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Quantum Dots (QD) – an Artifical Atom

- Interior of CdSe quantum dots

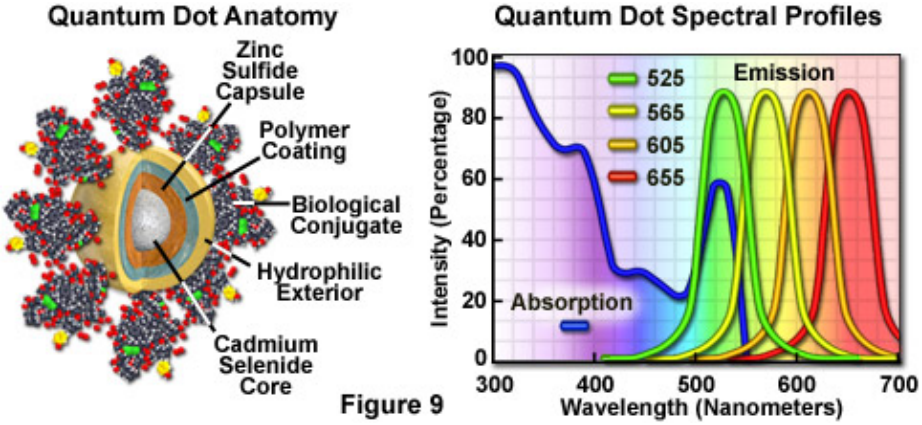
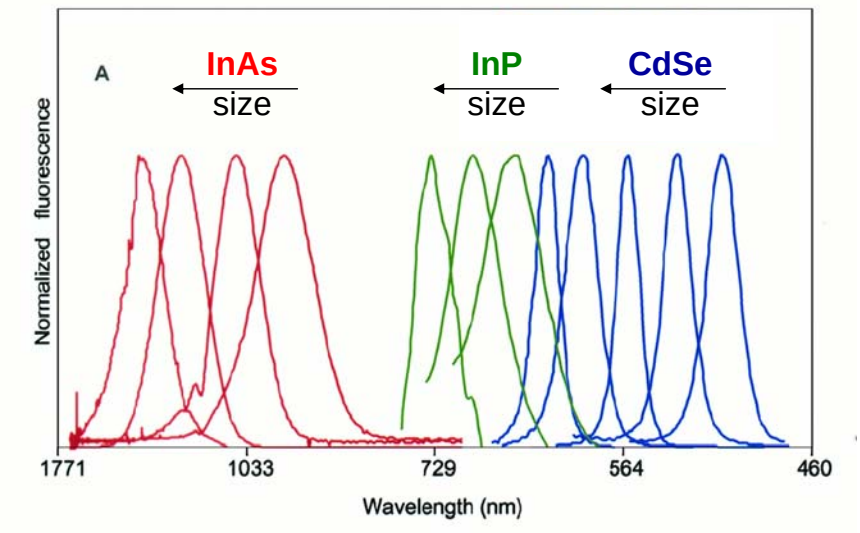
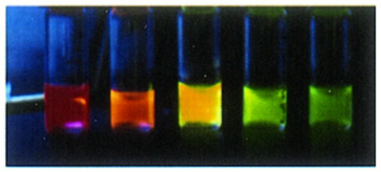


Figure 9

- Two disadvantages of CdSe QDs
  1. High toxicity – coating the QD's in ZnS to prevent the CdSe crystal from dissolving
  2. Insolubility in water – coating with a water-soluble hydrophilic polymer or silica

Fluorescence Spectra of Quantum Dots





Water-soluble silica-coated
Core (CdSe)-Shell (ZnS or CdS)

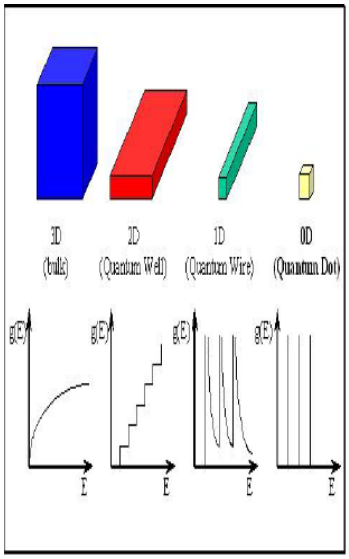
Characteristics of Quantum Dots (QD)

- Quantum dots are small devices that contain a tiny droplet of free electrons.
- Typical dimensions are between nanometers to a few microns.
- A quantum dot can have anything from a single electron to a collection of several thousands.
- The size ,shape and number of electrons can be precisely controlled.
- Just as in an atom, the energy levels are quantized due to the confinement of electrons.
- The 3D spatial confinement is observed in the quantum dots.
- Unlike atoms however, quantum dots can be easily connected to electrodes and are therefore excellent tools to study atomic-like properties…..

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History of Quantum Dots

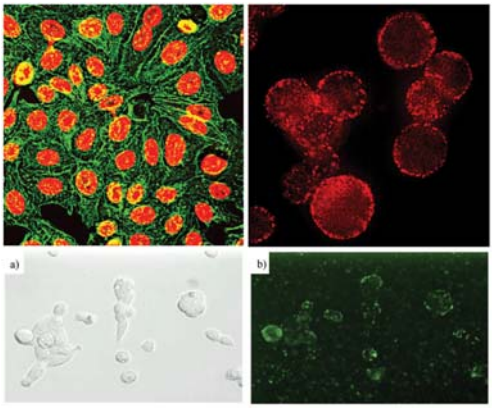
- In the 1970s the first low dimensional structures Quantum Wells were developed.
- 1D (quantum wires) and 0D (quantum dots) were subsequently developed.



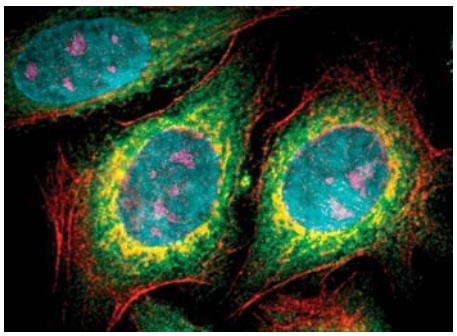
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Quantum Dots and Their Applications (I)

1. Quantum dots in biology and medicine
   - Advantages of QD over organic dye molecules: organic dye molecules bleach within a minute or two, while QD's do not bleach and they are also about 10- to 20-times as bright as commonly used organic fluorophores.



Parak et al., Nanotechnology **16**, R9 (2005)

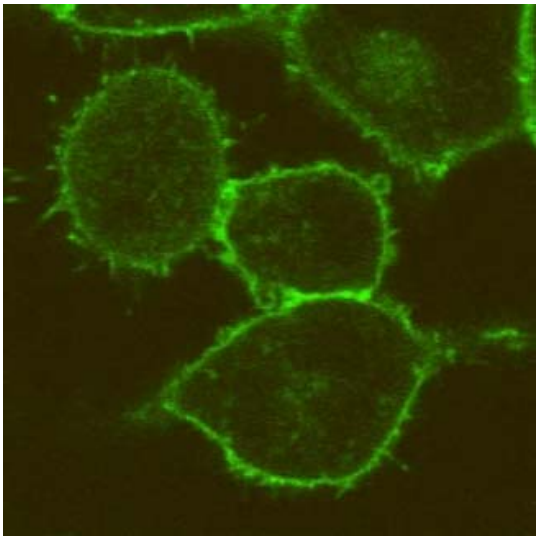


the nucleus is blue, a specific protein within the nucleus is pink, mitochondria look yellow, microtubules are green, and actin filaments are red . (Quantum Dot Corp.)

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Quantum Dots and Their Applications (II)

- Quantum dots to study living cells



Lidke et al., Nature Biotechnology **22**, 198 (2004)

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Quantum Dots and Their Applications (III)

- A limitation of the fluorescent dots is that visible light frequencies would only be useful near the surface of the skin.
  → QD's that emit near-infrared light.
  → Near infrared penetrates tissue much more efficiently than visible light.

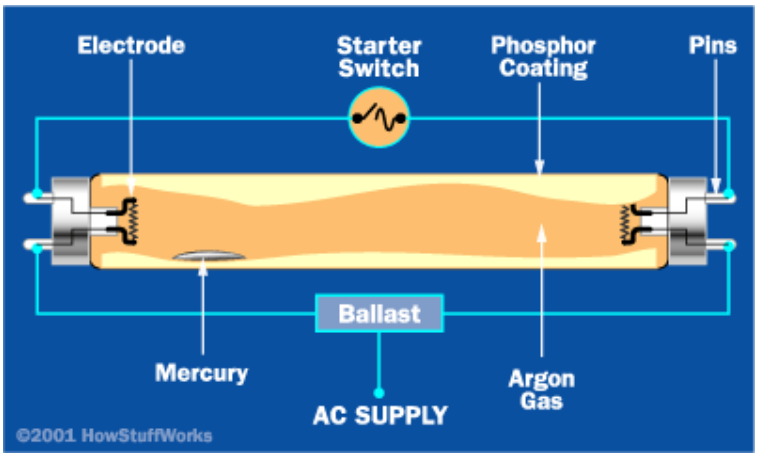


Kim et al., Nature Biotechnology **22**, 93 (2004)

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Quantum Dots and Their Applications (IV)

2. Quantum dots in Electronics

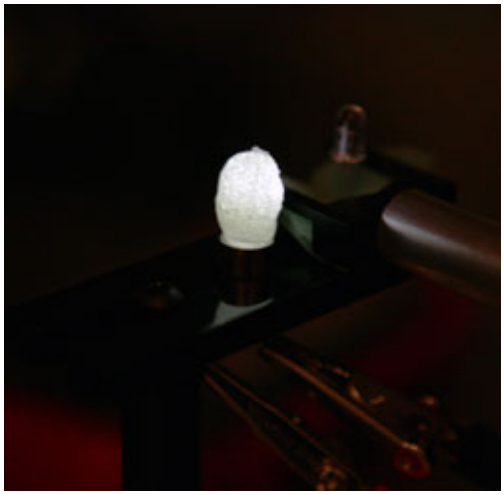


Conventional fluorescent lamp

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Quantum Dots and Their Applications (V)

2. Quantum dots in Electronics
    - Increasing the efficiency of fluorescent lighting.



QD fluorescent lamp